VIETNAM NATIONAL UNIVERSITY – HOCHIMINH CITY
INTERNATIONAL UNIVERSITY
DEPARTMENT OF BIOMEDICAL ENGINEERING

BME CAPSTONE DESIGN COURSE

PROJECT REPORT

Contact angle system

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Abstract

Contact angle measurement systems play a crucial role in laboratories for characterizing surface properties, wettability, and interfacial phenomena across various scientific and industrial applications. These systems are essential for research in materials science, biomedical engineering, and surface chemistry. Recognizing the importance of accurate and reliable contact angle measurements, this report addresses the limitations of an existing setup in Lab 406 and presents the design and implementation of an improved system. The original system in Lab 406 suffered from instability, an uneven sample stage incapable of tilting, and the absence of a syringe pump, compromising measurement accuracy and reproducibility. Our approach involved reconstructing the system using shaped aluminum profiles to ensure a stable foundation. We integrated a motorized syringe pump with a transmission system, allowing precise control of water droplet volume through button inputs. The sample stage was enhanced with a micro-adjuster, enabling tilt adjustments up to 10 degrees. Results show significantly improved stability and functionality of the overall system. The syringe pump and tilting mechanism performed as intended, addressing the primary issues of the original setup. However, one limitation remains: the sample stage lacks vertical adjustment capability. Future improvements could incorporate an additional micro-adjuster block to enable raising and lowering of the sample stage. This project demonstrates substantial enhancements to the contact angle measurement system, potentially improving the accuracy and reliability of surface characterization experiments in Lab 406. The improved system offers better control and precision, vital for advancing research and applications dependent on accurate surface property measurements.

Keywords: Contact angle, hydrophobicity, lab-made device, tilting

I. Introduction

A. Background

In the rapidly evolving field of biomedical engineering, the characterization of biomaterial surfaces plays a crucial role in advancing medical technologies and improving patient outcomes. The Biomedical Engineering Laboratory (Lab 406) at our institution specializes in biomaterials research, where surface wettability is a critical factor in determining material properties and predicting biological interactions. The accurate measurement of contact angles is fundamental to assessing surface wettability, which directly influences protein adsorption, cell adhesion, and overall biocompatibility of materials used in medical implants, drug delivery systems, and tissue engineering scaffolds. However, the existing contact angle measurement system in Lab 406 has been plagued by instability issues and lacks essential features, severely limiting the quality and reliability of research outcomes. The current system's inconsistent performance, coupled with its inability to tilt the sample stage or control droplet volume precisely, has hindered researchers' ability to conduct accurate and reproducible experiments. These limitations have potentially impeded progress in developing novel biomaterials and optimizing existing ones for various medical applications. To address these critical issues, our team has developed an enhanced contact angle measurement system that not only rectifies the shortcomings of the old device but also introduces advanced features to meet the demanding requirements of modern biomaterials research. Our innovative design incorporates a stable foundation using shaped aluminum profiles, ensuring consistent measurements across various sample types. The integration of a motorized syringe pump with precise volume control allows researchers to deposit droplets with unprecedented accuracy, a feature crucial for studying the dynamic interactions between biomaterials and physiological fluids. Furthermore, the addition of a tiltable sample stage, capable of adjustments up to 10 degrees, enables the investigation of surface properties under different orientations, mimicking real-world conditions more closely. By addressing these key limitations, our upgraded contact angle system empowers researchers in Lab 406 to conduct more sophisticated experiments, generate more reliable data, and ultimately accelerate the development of advanced biomaterials for a wide range of medical applications.

B. Project objectives and motivation

Project goals:

The primary goal of this project is to develop an advanced contact angle measurement system that addresses the critical limitations of the existing equipment in the Biomedical Engineering Laboratory (Lab 406), while simultaneously advancing the capabilities of biomaterials research. Our aim is to create a compact, portable, and highly accurate device that incorporates precise droplet control, a tiltable sample stage, and a user-friendly interface. This enhanced system will enable researchers to conduct more sophisticated experiments, generate more reliable data, and ultimately accelerate the development of advanced biomaterials for a wide range of medical applications. By integrating features such as a stable aluminum frame, motorized syringe pump with precise volume control, and a camera with adjustable positioning, we seek to significantly improve the consistency and reproducibility of contact angle measurements.

Motivations:

The motivation behind this project stems from the crucial role that surface wettability plays in determining the properties and biological interactions of biomaterials. In the rapidly evolving field of biomedical engineering, accurate characterization of material surfaces is essential for advancing medical technologies and improving patient outcomes. The current system's instability and lack of essential features have hindered progress in developing novel biomaterials and optimizing existing ones for various medical applications. By creating a more versatile and reliable contact angle measurement system, we aim to empower researchers in Lab 406 to explore new avenues in biomaterials research, potentially leading to breakthroughs in areas such as medical implants, drug delivery systems, and tissue engineering scaffolds. This project not only addresses an immediate need within our laboratory but also contributes to the broader field of biomedical engineering by providing a tool that can enhance the quality and scope of surface characterization studies.

C. Literature Review

1. Existing methods

The quantification of contact angle (θ) represents a fundamental and widely employed technique for the characterization of surface properties across diverse applications, encompassing superhydrophobic surfaces, inkjet printing, and spray cooling technologies. This parameter, θ, serves as a macroscopic indicator of the complex interfacial interactions between a liquid and a solid substrate, elucidating surface chemistry, topography, and capillary forces at micro- and nanoscales (Lubbers et al., 2014).

The conceptual framework of contact angle for a liquid droplet on a solid surface was initially posited by Thomas Young in 1805. The eponymous equation, $\gamma_{ls} + \gamma_{lg} \cos \theta_e = \gamma_{sg}$, derived through force balance analysis, correlates the equilibrium contact angle (θ_e) with the surface tensions (γ) operating at the three-phase contact line, where liquid (l), gas (g), and solid (s) interfaces converge (Kung et al., 2019). This relationship can also be derived via Gibbs free energy minimization, employing a thermodynamic approach.

Recent technological advancements, particularly the proliferation of cost-effective digital imaging systems and sophisticated image analysis software, have rendered contact angle measurements increasingly accessible to research laboratories. Contemporary goniometric apparatus for θ measurement typically comprises an adjustable sample stage, illumination source, motorized liquid dispensing system, and high-resolution imaging device for capturing static images or dynamic video sequences.

Commercial systems often incorporate software-controlled droplet deposition and image acquisition, interfacing with a computer for data processing. Imaging devices vary in sensor technology (CCD or CMOS) and resolution, typically ranging from 2 to 50 μm/px (equivalent to 20-500 ppm, pixels per millimeter) (Wang et al., 2014). Portable systems leveraging smartphone technology have also emerged, offering rapid and automated droplet analysis at reduced cost.

Post-image acquisition, contact angle determination involves several methodological approaches such as Secant method. θ is measured by fitting a tangent line to the droplet profile at the contact point (CP). Next is fitting method which is the most prevalent technique in conventional and commercial software, employing the Young-Laplace equation, circular, ellipsoidal, or polynomial functions to model the droplet profile proximal to the contact point. The last method is Mask, which utilizes convolution between a mask matrix and the image to calculate local slopes of the droplet profile.

Image analysis for contact angle evaluation is typically performed using proprietary software bundled with commercial equipment. Alternatively, research groups often develop custom algorithms. Notable open-source alternatives include DropSnake and LB-ADSA, which are available as plugins for the ImageJ platform.

2. Existing devices/applications on markets

Contact angle measurement systems play a crucial role in surface science and biomaterials research, providing essential data on surface properties and fluid-surface interactions. Current methodologies in the field include the sessile drop method, Wilhelmy plate method, and captive bubble technique. Each of these approaches offers distinct advantages and limitations, catering to different research needs and sample types. The sessile drop method, while simple and widely applicable, is limited to static measurements and susceptible to evaporation errors. The Wilhelmy plate method excels in providing dynamic measurements and is suitable for powders and fibers, but requires specific sample preparation. The captive bubble method is particularly useful for hydrophilic surfaces and minimizes evaporation issues, yet it involves a more complex setup and is limited in sample compatibility.

The market for contact angle measurement systems is dominated by several key players, each offering unique features and capabilities. The DataPhysics OCA Series, originating from Germany, is known for its high-precision optical systems with resolutions up to 0.01° and modular design allowing for customization. However, its high price point and complex user interface may present barriers for some researchers.

Figure 1. OCA Optical contact angle measuring and contour analysis systems (OCA Optical Contact Angle Measuring and Contour Analysis Systems - DataPhysics Instruments, n.d.)

The KRÜSS DSA100, also from Germany, boasts a high-speed camera capable of up to 3000 fps and robust design suitable for both industrial and academic settings. While it offers comprehensive features, its premium pricing and substantial size may be limiting factors.

Figure 2. Drop Shape Analyzer – DSA100 (Drop Shape Analyzer-DSA100E, n.d.)

The Biolin Scientific Attension Theta, hailing from Finland, presents a more compact design with a user-friendly touchscreen interface, striking a balance between features and price. However, it may offer lower frame rates compared to high-end systems and limited customization options.

Figure 3. Attension Theta Optical Tensiometers – Biolin (Optical Tensiometers | Attension, n.d.)

American manufacturers also contribute significantly to the market. The ramé-hart Model 250 is known for its modular design and robust construction, offering a relatively affordable base model. However, basic models may lack some advanced features, and the software interface can appear dated.

Figure 4. ramé-hart Goniometer / Tensiometer Model 250 (Ramé-Hart Surface Science Instruments, n.d.)

The First Ten Angstroms FTÅ200 provides a compact, benchtop design with good optical resolution and specialization in high-temperature measurements. Its limitations include restricted automation features in base models and a potentially steeper software learning curve.

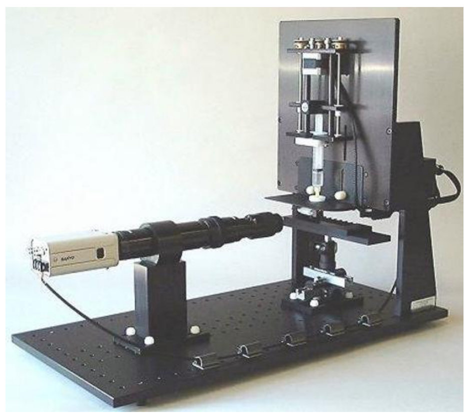


Figure 5. FTÅ200 - First Ten Angstroms (FTÅ200, n.d.)

From Japan, the Kyowa Interface Science DMs-400 offers high-precision measurements and fully automated operation, with specialized models for specific industries. However, its higher price point and potentially overspecialized features for general research use may limit its appeal in some markets.

Figure 6. DMs-401 - Kyowa Interface Science (DMs-401：Kyowa Interface Science, n.d.)

The clear dominance of European and American manufacturers reflects not only the historical development of surface science in these regions but also their strong economic and technological infrastructures. This concentration of expertise has led to a competitive market dynamic, with these manufacturers investing heavily in research and development to maintain their market positions.

The industry's shift towards modular, customizable designs represents a strategic response to diverse research needs and budget constraints. This trend allows laboratories to make initial investments in basic systems and gradually upgrade as funding becomes available or research needs evolve. For instance, a basic contact angle measurement system might cost around $15,000, with modular additions for features like temperature control or automated dispensing ranging from $5,000 to $20,000 each. This approach enables labs to spread costs over time and align equipment capabilities with specific grant funding or project requirements.

The increasing focus on automation and high-speed imaging is driven by the economic imperative to maximize research output and efficiency. While these advanced systems command premium prices, often exceeding $50,000, they can significantly reduce labor costs and increase throughput. For example, an automated system might process 100 samples in the time a manual system handles 10, potentially justifying the higher initial investment for labs with high-volume needs.

Compact, benchtop models gaining popularity for routine measurements reflect a market segment catering to budget-conscious labs and industry quality control applications. These systems, typically priced between $10,000 and $25,000, offer a cost-effective solution for basic research and teaching laboratories. Their growing adoption is partly driven by the expansion of materials science and nanotechnology research in smaller institutions and developing countries, where space and budget constraints are significant factors.

Despite these advancements, common limitations persist, each with economic ramifications. Operational complexity requiring significant training translates to hidden costs in terms of staff time and potential measurement errors. High-end systems often necessitate specialized training courses, which can cost $1,000 to $3,000 per person, not including travel expenses. Some manufacturers offer on-site training as part of the purchase package, typically valued at $5,000 to $10,000, highlighting the significant investment required to fully utilize these instruments. High costs continue to limit accessibility, with top-tier systems often priced between $40,000 and $100,000. This cost barrier can lead to resource inequality among research institutions, potentially impacting the competitiveness of smaller labs or those in less-funded disciplines. To address this, some manufacturers offer leasing options or refurbished systems, which can reduce initial outlays by 30-50%. Large footprints in some systems, requiring dedicated lab space, represent an often-overlooked economic factor. In regions with high real estate costs, the space requirement for a large system can translate to thousands of dollars annually in indirect expenses. This has driven demand for more compact designs, which can offer significant savings in space-constrained environments. Overspecialization of features in high-end systems can lead to inefficient capital allocation, with labs paying for capabilities they may rarely use. This trend has sparked a market for pre-owned and refurbished equipment, where labs can acquire high-end systems at 40-60% of the original cost, albeit with potential trade-offs in warranty coverage and future support. Limited flexibility for custom research applications often necessitates in-house modifications or the development of custom apparatus. While this can drive innovation, it also represents a significant investment of time and resources. Some labs report spending $10,000 to $50,000 on custom modifications to commercial systems, highlighting the economic impact of this limitation.

The economic landscape of contact angle measurement systems also reflects broader trends in scientific funding and research priorities. Government grants and industry partnerships play a crucial role in equipment acquisition, with funding bodies increasingly emphasizing shared instrumentation grants to maximize resource utilization across multiple research groups or institutions. Furthermore, the total cost of ownership, including maintenance contracts (typically 10-15% of the purchase price annually), software updates, and consumables, can significantly impact a lab’s operational budget over the lifespan of the equipment. Some manufacturers are addressing this by offering all-inclusive service packages or pay-per-use models for advanced features, allowing labs to better manage long-term costs.

In response to these market gaps, our innovative design proposal for a contact angle measurement system addresses key limitations in current offerings. The system incorporates a stable foundation utilizing shaped aluminum profiles, ensuring measurement consistency across diverse sample types. A standout feature is the integration of a motorized syringe pump with precise volume control, enabling accurate droplet deposition and facilitating the study of dynamic interactions between biomaterials and physiological fluids. Additionally, the inclusion of a tiltable sample stage, adjustable up to 10 degrees, allows for the investigation of surface properties under various orientations, more closely simulating real-world conditions. These features collectively enhance measurement precision and expand the scope of potential research applications, particularly in the field of biomaterials.

Looking towards future developments, several enhancements could further advance the system’s capabilities. These include the integration of multi-fluid dispensing capabilities, a temperature-controlled sample stage, an environmental chamber for atmosphere control, high-speed imaging integration, machine learning algorithms for data analysis, and an open-source software platform. By addressing significant gaps in current market offerings and combining precision, flexibility, and specialized features tailored to biomaterials research, this innovative design has the potential to advance surface science studies while offering improved accessibility to researchers. Its unique features, particularly the tiltable stage and high-precision syringe pump, position it as a valuable tool for investigating complex surface-fluid interactions in physiologically relevant conditions, potentially opening new avenues in biomaterials research and development.

D. Design requirements

The proposed contact angle measurement system is designed to be compact and portable, with dimensions of 50 cm x 30 cm x 20 cm (L x W x H) and a weight of approximately 2 kg. It features convenient carrying handles on both sides and a stable base constructed with shaped aluminum framing. The system incorporates a user-friendly interface with an LCD screen and dedicated buttons for various functions, including droplet size selection (5, 10, and 20 microliters), a "drop" button for droplet release, and controls for the motorized syringe pump’s z-axis movement. A high-resolution camera, mounted on a sliding mechanism for x-axis adjustment, captures images for accurate contact angle measurements. The sample stage is manually adjustable, allowing tilting up to 10 degrees in 1-degree increments. This design prioritizes precision, with target measurement accuracy of ±0.1 degree or better, while maintaining ease of use. The system’s modular construction facilitates potential future upgrades and ensures easy maintenance. Power consumption is optimized for standard laboratory sources, and safety features include an enclosed droplet dispensing area. This integrated design balances portability, accuracy, and versatility, making it suitable for a wide range of contact angle measurement applications in research and industrial settings.

II. Methodology

A. Design Overview

This process is essential to ensure that the final product meets the needs of the Biomedical Engineering Laboratory (Lab 406) while addressing the limitations of the existing system.

In our design selection process, we will prioritize features that directly contribute to measurement accuracy, ease of use, and versatility in biomaterials research. Key considerations will include the stability of the system, precision of droplet control, adjustability of the sample stage, and overall portability. We will evaluate design elements based on their ability to improve measurement consistency, facilitate a wider range of experiments, and enhance user experience.

Rationales for selection will include factors such as compatibility with the specified size and weight constraints, potential for improving measurement accuracy, ability to integrate with existing laboratory workflows, and cost-effectiveness. We will favor designs that incorporate modular components, allowing for future upgrades and customization. Additionally, we will consider the manufacturability and maintainability of each design element.

Conversely, rationales for rejection will involve identifying design elements that fail to meet our specified requirements, introduce unnecessary complexity, or potentially compromise the system’s reliability. We will critically assess each feature’s contribution to the overall functionality of the system, rejecting those that add bulk without proportional benefits or that might introduce new sources of measurement error. Through this systematic approach to design selection and rejection, we aim to develop a contact angle measurement system that not only meets the immediate needs of Lab 406 but also sets a new standard for precision and versatility in biomaterials surface characterization.

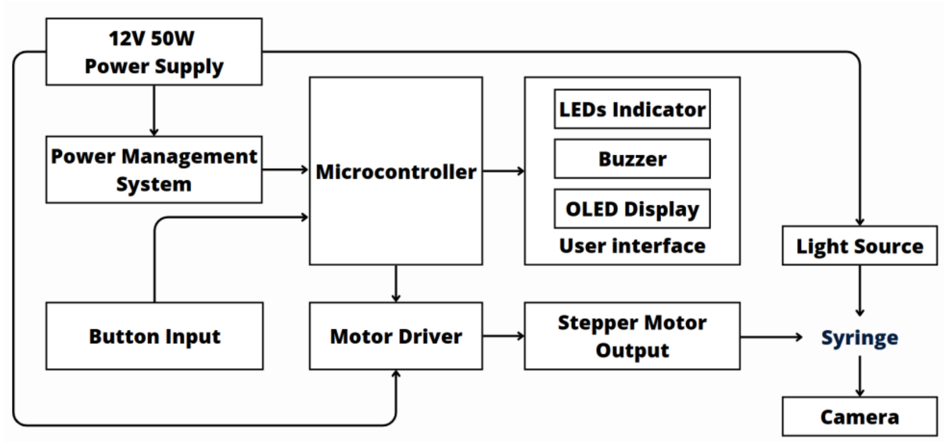


Figure 7. Block diagram of the contact angle system

B. Mechanical Description

Decision matrixes of mechanical design

Table 1. Decision matrix of CAD softwares

Software	Prices	Ease of Use	Student Friendly	Learning Material	Total Score
Fusion 360	**4**	**5**	**5**	**5**	**19**
SolidWorks	3	4	4	4	15
AutoCAD	3	4	4	4	15
FreeCAD	5	3	4	3	15
Rhino	3	4	3	4	14

- **Prices**: (1-5, 5 being the most affordable)
- **Ease of Use**: (1-5, 5 being the easiest)
- **Student Friendly**: (1-5, 5 being the most student-friendly)
- **Learning Material**: (1-5, 5 having the most learning resources available)

C. Electrical Description

1. Decision matrixes for selecting hardware components

The system's electrical components consist of 4 main parts: the microcontroller, the motor, the light source and the user interface. For precise control of the motor displacement to pump a minute droplet volume, the best option would be to choose the stepper motor with the smallest step angle. A 1.8 step angle motor was selected to accomplish the system requirement because of its high availability. The decision matrix for the motor driver is elaborated below.

1.1. Decision matrix for selecting the motor driver

	A4988	TB6600	DRV8825
Cost	5 (54.000 VND)	2 (290.000)	4 (70.000)
Durability x 2	3	5	3
Power	3 (max. 2A)	5 (max. 5A)	4 (max. 2.5A)
Microstepping	4 (max. 1/16)	4 (max. 1/16)	5 (max. 1/32)
Total	18	21	19

Figure 8. Decision matrix for selecting the motor driver

The decision matrix shows the most prevalent micro-stepping motor drivers on the market. The micro-stepping control method can achieve high resolution and produce smooth movement, making it a common method for controlling a stepper motor. The A4988 and DRV8825 motor drivers are popular among hobbyists due to their low cost and small size, but they are more susceptible to high temperatures, which reduces their durability. The TB6600 driver is frequently used in industry thanks to its good performance and durability. Therefore, selecting this driver ensures accurate outcome and prolonged use, making it suitable for laboratory applications.

1.2. Decision matrix for selecting the light source

	12VDC 10W	220VAC 10W
Size	5 (5x5 cm)	3 (12x12 cm)
Availablility	3	5
Safety x 2	5	3
Easy for integration	5	3
Total	18	14

Figure 9. Decision matrix for selecting the light source

The light source should be chosen with adequate light intensity to highlight the droplet surface morphology. After intensive experimentation, light sources with 10W power have been proven to produce the best contrast while maintaining low power consumption. Although 220V AC-voltage-supply light sources are prevalent in the market due to their common use as ceiling lights, they are not suitable to our system. Therefore, we opted for 12V DC-voltage-supply light sources, even though those with pre-designed housing and heat sinks are more challenging to find.

1.3. Decision matrix for selecting the microcontroller

	ESP32 DevKitC	Arduino Uno	STM32F103
Cost	3 (118.000VND)	4 (94.000VND)	5 (53.000VND)
Familiarity x 2	4	5	2
Number of GPIO pins	5 (36)	3 (14)	5 (36)
Size	4	3	5
Functionality	5 (32-bit, Wifi+Bluetooth)	3 (8-bit, limited timer configuration)	4 (32-bit)
Total	25	23	23

Figure 10. Decision matrix for selecting the microcontroller

Them less suitable for our needs. Therefore, we opt for the ESP32 microcontroller to act as the head of our system, with The Arduino Uno is well-known among makers for its simple and beginner-friendly approach. For this simple project, the Arduino Uno is sufficient as no advanced ESP32 features are required. However, as the project progresses, the use of ESP32 becomes more prospective. The main reason to switch to the ESP32 is that the ESP32 Arduino Core allows versatile adjustments of the MCU's timers, which is a crucial feature for controlling the pulse supplied to the motor driver. The ESP32 Arduino Core provides timer's libraries with useful functions like timerInit(), timerStart()…, allowing programmers to easily set up and manage timers. In contrast, controlling the Arduino Uno's timers requires low-level register programming, which is undesirable as it adds unnecessary complexity.

In addition to the two mainstream user-friendly microcontrollers, the industrial STM32 microcontrollers are worth considering since they are powerful yet low-cost. However, they use unfamiliar IDEs like STM32CubeIDE or Keil uVision, making abundant functionality while remaining familiar enough for us to work with.

2. <u>Electrical schematics</u>

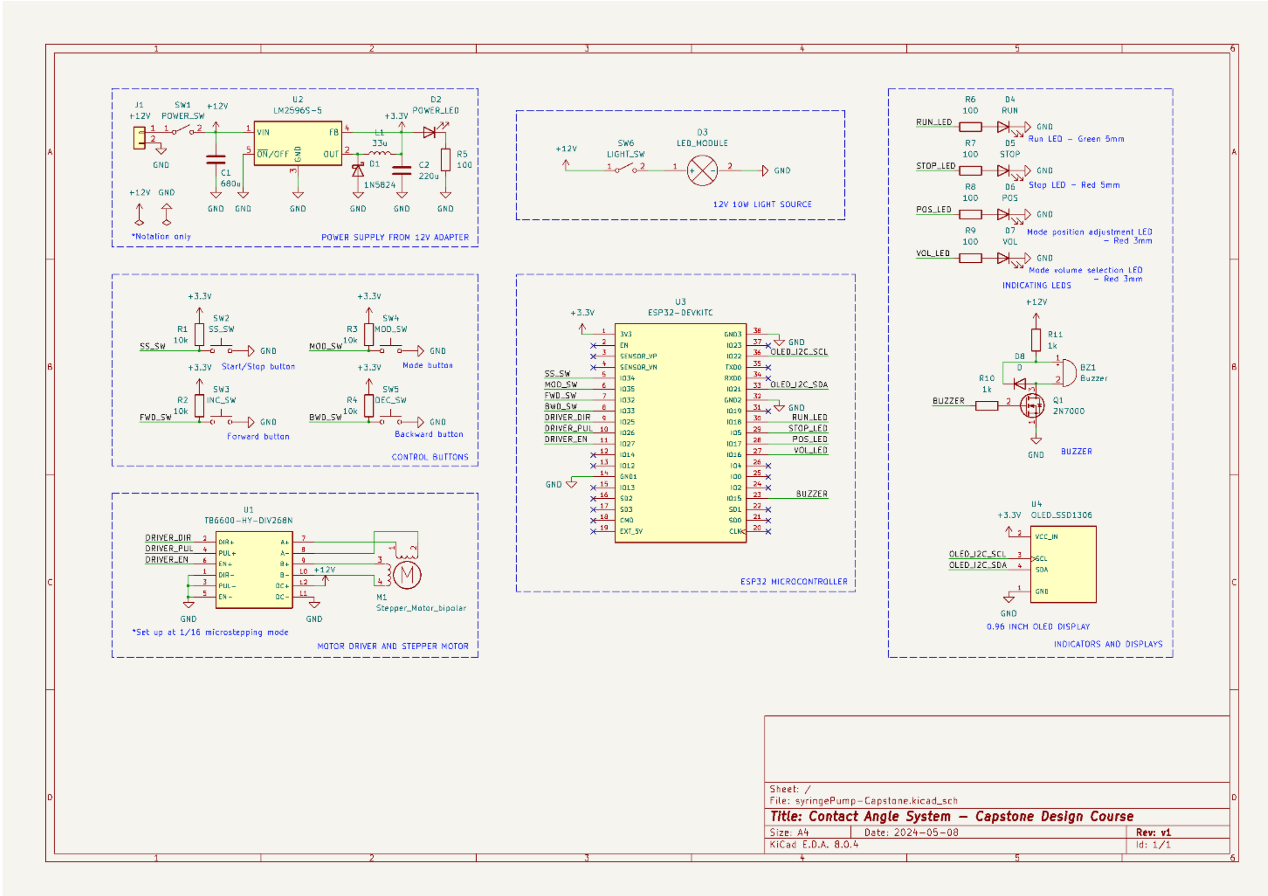


Figure 11. Electrical schematics of the system

Figure 11 shows the complete schematics of the system. The device's design was created using the Kicad electronic design automation (EDA) platform. Kicad is a free and open-source EDA software suite that provides an integrated environment for tasks like schematic design, PCB layout, manufacturing file viewing, and engineering calculations. The components within the schematics can be divided into six groups. The descriptions of some elements are provided below.

2.1. Power supply

A 12V adapter with 50W power supply was utilized to convert AC line voltage into DC voltage. Then the circuit used a step-down voltage regulator LM2596 to provide a constant 3.3V for the ESP32 microcontroller and the user interface. LM2596 is a popular switching regulator that can drive a load of 3A, with more power and higher efficiency than the common 7805 IC that can deliver an output of only 1.5A current. The power LED will turn on when the power switch is closed.

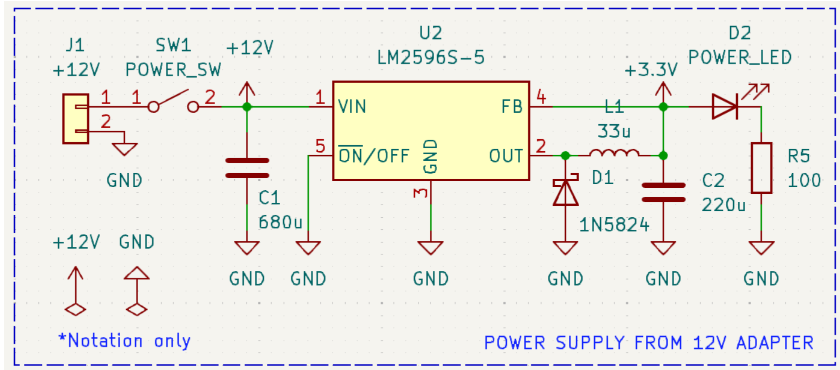


Figure 12. Schematics of the power supply

2.2. Motor and motor driver

The TB6600HC is a professional bipolar stepper motor driver that is commonly used to control stepper motors. The driver generates pulse-width-modulated signals to control the current supply to each phase of the stepper motor. It supports power input from 9V to 42V DC and offers microstepping ability of up to 32 microsteps per step. Here, we used the commercial package with pre-designed circuit for the TB6600. To set up the microstepping mode of 1/16, three switches S1, S2, and S3 on the side of the package are toggled in a certain sequence, with S1 and S2 are off when S3 is on. The current was set based on the motor used. For our NEMA 17 stepper motor, the suitable current supplied was 1A. Exceeding or below this value would stop the motor from running.

Figure 13. Stepper motor driver TB6600HC Pro diagram

The signal was driven by the microcontroller output from the ESP32. There are three logical inputs from the microcontroller, the Enable (EN) input, Direction (DIR) input, and the Clock (PUL) input. To turn on the current output, meaning to run the motor, the EN input is configured high. To stop the motor, the EN input is configured low. Whenever the PUL input has edge transition from Low to High, the output current will change; the motor then achieves one microsteps subsequently. Depending on the state of the DIR input, the waveform of the output current in each phase would change, making the motor rotate either clockwise (CW) or counterclockwise (CCW). Figure 14 describes how the PUL input and EN input regulate the motor operation, while Appendix A demonstrates the current outputs in each phase when the DIR input is low (CW mode) and when it is high (CCW mode). The magnitude of these currents is defined by the switch S4, S5, S6 settings shown in Figure 13.

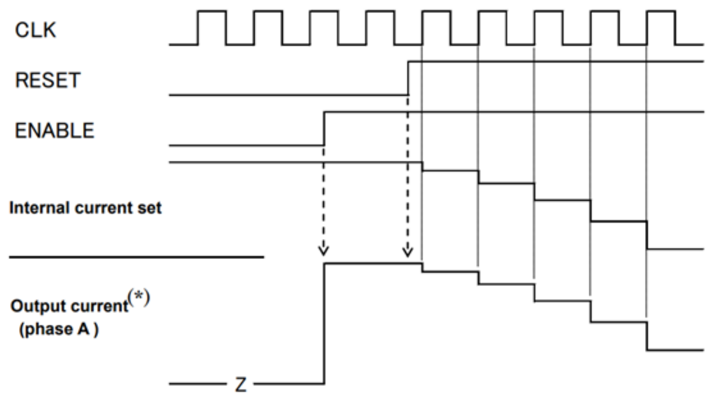


Figure 14. Motor driver timing diagram

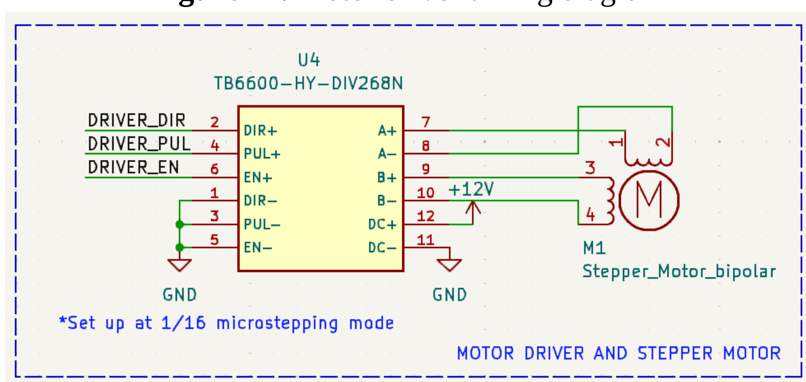


Figure 15. Schematics of the motor and motor driver

2.3. Light source

The light source is a recycling electric bike front light comes in compact module with mounted heat sink. Figure 17 shows the actual image of this light source. The light source running on 12V power supply was controlled manually by a mechanical switch attached to the side of the device.

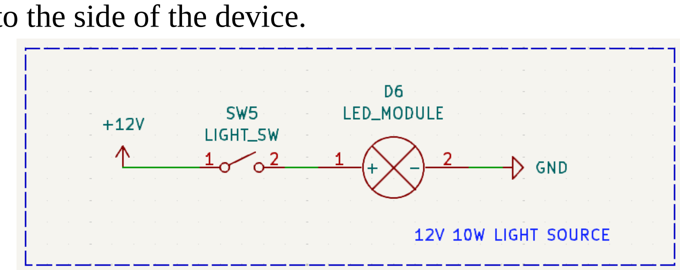


Figure 16. Schematics of the light source

Figure 17. Image of the recycled 12V light source

2.4. User interface

The user interface is where interactions between users and the system occur. The users control the device by using four buttons, namely Mode button, Start/Stop button, Backward button and Forward button. Those buttons are wired in pull-up configuration, meaning their default input state is high.

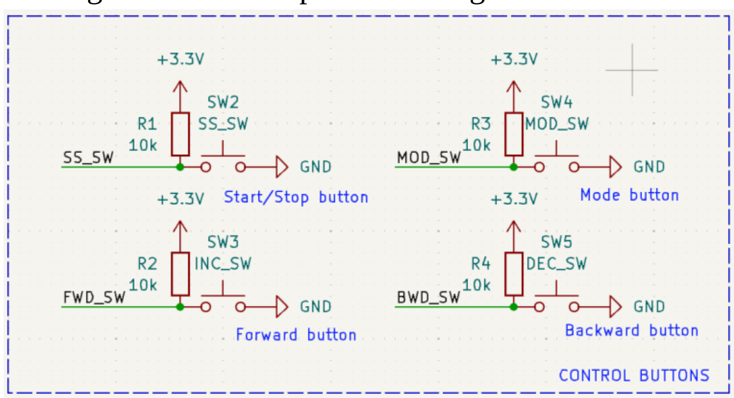


Figure 18. Schematics of the buttons

To inform users about the system status, four LEDs and an OLED display are utilized. The green light is on meaning the motor is running while the red light means the motor stops. The OLED display shows the current mode and the preset volume. When the current mode is *Position adjustment*, the LED D4 is lit up. Similarly, when the current mode is *Volume selection*, the LED D5 turns on. To make the interaction become more immersive, a buzzer will emit sound whenever users press a button.

The OLED display communicates with the ESP32 through I2C protocol. The protocol requires two wires to make communication. The SDA wire (Serial Data) is used for sending and receiving data while the SCL wire (Serial Clock) carries the clock signal. The Arduino IDE provides the Adafruit SSD1306 library to support the communication between ESP32 and the OLED display.

Figure 19 shows the wiring of the LEDs, buzzer and the OLED display.

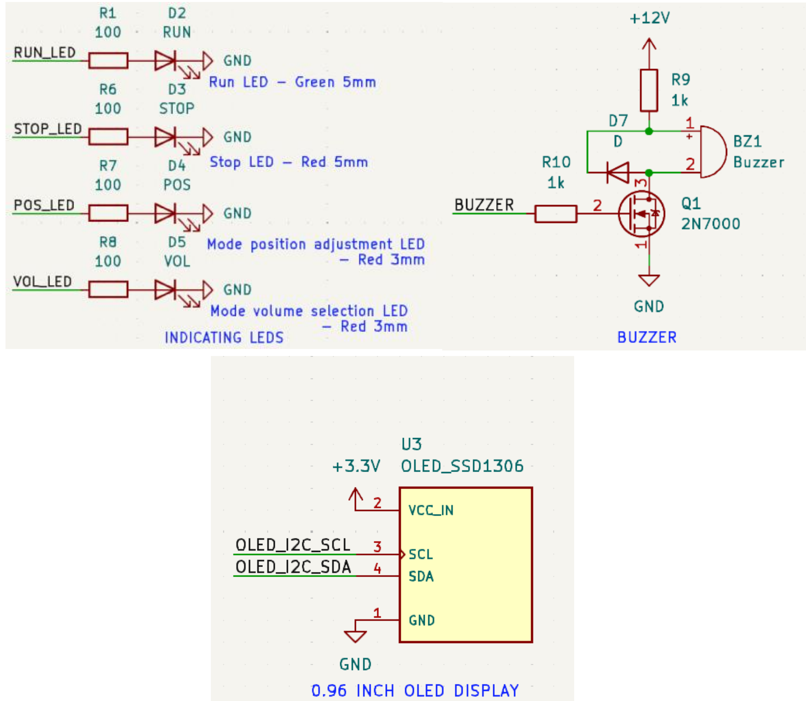


Figure 19. Schematics of the indicators, including LEDs and buzzer, and the OLED display

D. Software Description

1. Decision matrix for selecting the ESP32 development framework

	ESP Arduino Core	ESP-IDF
Learning curve x2	5	3
Feature support	3	5 (low-power mode, OTA updates, FreeRTOS)
Programming Language	4 (Simplified C++)	5 (C/C++)
Total	17	16

Figure 20. Decision matrix for selecting the ESP32 development framework

To program the ESP32, there are two frameworks to choose from: the ESP Arduino Core and ESP-IDF (ESP IoT Development Framework). Development frameworks include all the necessary software libraries and source code for building software that runs on the ESP32. The ESP Arduino Core is specifically designed to simplify the programming process by leveraging the familiar Arduino platform, making it easier for programmers who are familiar with Arduino IDE to start programming the ESP32. However, being a simplified version, this framework lacks access to many advanced features of the ESP32 such as low-power mode, over-the-air (OTA) updates and real-time operating system (FreeRTOS). Additionally, the simplified C++ language of the Arduino IDE, while beneficial for beginners, may limit the language's full capabilities. Since our system's initial requirements do not necessitate incorporating advanced features like Wifi and FreeRTOS, the selection is made with ESP Arduino Core as it streamlines the development process, making it less time consuming when programming ESP32.

2. State machine diagram of the software

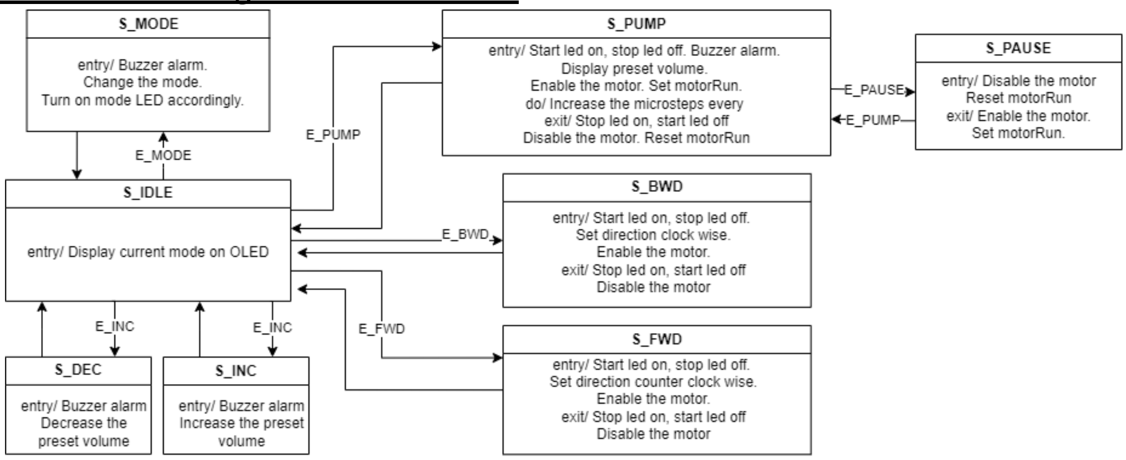


Figure 21. State machine diagram of the software

The code for the whole system is based on state machine programming, which is an abstract model that crafts a system into multiple different states. The system can be in one state at a time. Each state specifies the actions performed and which state to transition to for a given input.

The system received input through four buttons, namely Mode, Backward, Forward and Start/stop buttons. Throughout the operation, the system continuously polls the button pins. An event is triggered if a button is pressed, and the related condition is met.

Table 2 describes how the system responds to the button inputs and returns the event accordingly.

Table 2. State machine programming: Event triggered based on button inputs

Press button	Condition	Trigger event	Explanation
Mode button	None	E_MODE	Event to change the currentMode
Backward button	currentMode == VOLUME	E_DEC	Event to decrease the preset volume
	currentMode == POSITION	E_BWD	Event to move the plunger backward
Forward button	currentMode == VOLUME	E_INC	Event to increase the preset volume
	currentMode == POSITION	E_FWD	Event to move the plunger forward
Start/Stop button	motorRun == 1	E_PAUSE	Event to pause the pumping
	motorRun == 0	E_PUMP	Event to start the pumping

Figure 20 illustrates how the current state specifies its transition and the action performed within each state. After the event is triggered, the system will check its current state. If the current state defines transition given by the event, then new transition happens; the system exits the current state and entries the next state. The actions fall into 3 types: entry, do and exit actions. Entry and exit actions are performed once the system is entering or existing the state. The do action is performed in a loop until the next transition occurs.

3. Flowchart of the software

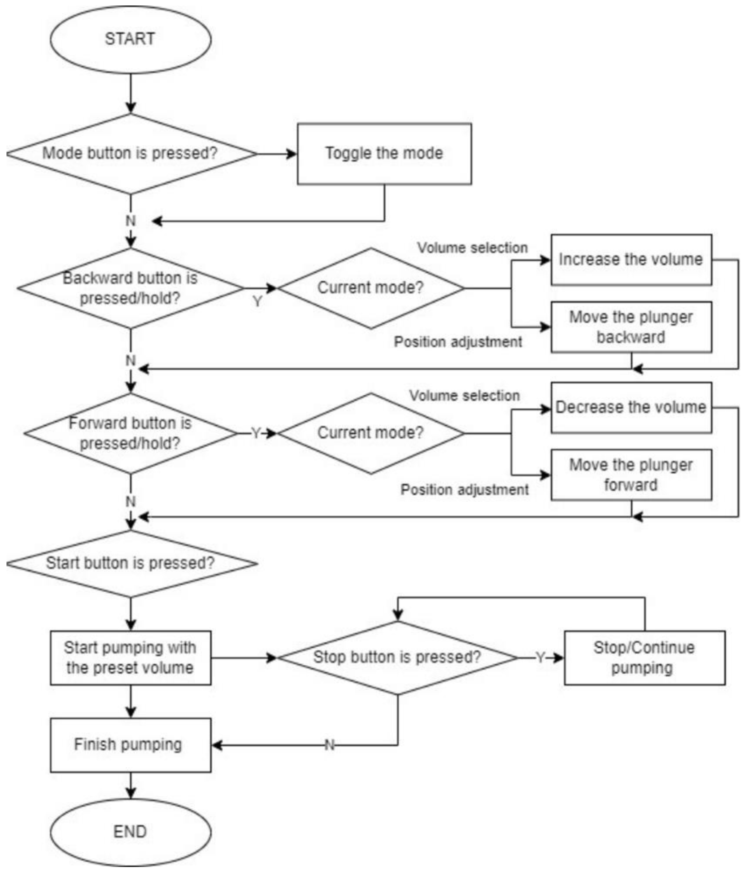


Figure 22. Flow chart of the software

Figure 22 is a flow chart demonstrating how to use the device. There are two modes of the system: (1) Volume selection, and (2) Position adjustment. The Mode button is used to toggle between the two modes. Based on the mode, the Backward and Forward button will have different functions.

Typically, a user will use the Mode button to select the Position adjustment mode, and then use the Forward and Backward buttons to move the plunger up or down. After the syringe filled with water is inserted into the plunger holder and all the air bubbles inside the syringe are removed, the user now changes the mode into Volume selection. The Backward and Forward buttons now function to decrease or increase the preset volume. After the volume is selected, users now press the Start/stop button to start pumping. In case the user wants to abruptly stop the pumping process, they can repress the Start/stop button to halt the system operation. Select the Start/stop button once more to continue pumping the residual volume. The system will return to its idle state and wait for input when the pumping is finished.

4. Software algorithm

4.1 Pumping speed algorithm

Before proceeding with the algorithm, some of the important motor parameters should be mentioned. Each stepper motor is given with a step angle. For example, a 1.8-degree stepper motor has a step angle of 1.8°, meaning that to complete one revolution, the motor should rotate $360°/1.8° = 200[steps/rev]$. Based on the controlling mode of the stepper motor driver, one step of the motor can be divided into many microsteps such as $16\ [usteps/step]$, refer to table… The multiplication of these two parameters resulted in a new parameter, called (*number of) microsteps per revolution [usteps/rev]*, which is useful to calculate other control parameters.

To calculate the required time to trigger the timer which generates pulses that are supplied to the motor driver, first we need to find the desired pumping speed based on a given flow rate. After experimenting, we found that the flow rate of 50 uL/s was suitable to drive the motor in backward and forward direction. Faster speed resulted in low torque, and slower speed making the system aggressively vibrate. The formula for finding the pumping speed based on flow rate is:

$$pumping\ speed\ \left[\frac{mm}{s}\right] = \frac{flow\ rate\ [uL/s]}{area[mm^2]}$$

Since our system uses a 1cc syringe with a diameter of 4.78 mm.

We set the flow rate for one droplet manually, so that each droplet was pumped at the exact same time of 1 second. For example, for 5 uL droplet, the flow rate was 5 uL/s. Similarly, the flow rate for 10 uL droplet would be 10 uL/s.

4.2 Timer alarm algorithm

The timer was set with a frequency of 1MHz, meaning that …

The required time for the motor to rotate one revolution is:

$$time\ one\ revolution\ [s/rev] = \frac{screw\ pitch\ [mm/rev]}{pumping\ speed[mm/s]}$$

Therefore, the time to generate one pulse for the motor driver is:

$$time\ one\ pulse\ [us/pulse] = \frac{time\ one\ revolution\ [s/rev] \times 1000000[us/s]}{microstep\ per\ revolution\ [usteps/rev] \times 2[pulse/ustep]}$$

This means the timer will trigger after $time\ one\ pulse$ microseconds to achieve the desired flow rate.

4.3 Microsteps algorithm

The relationship between the droplet volume and the number of microsteps required is given by:

$$number\ of\ microsteps\ [usteps] = \frac{volume[uL] \times microstep\ per\ revolution\ [usteps/rev]}{area[mm^2] \times screw\ pitch[mm/rev]}$$

Each time the timer is triggered, the microstep will be increased by half a value. The system will disable the motor when the number of microsteps achieve the required based on the droplet volume.

E. Final Prototype

a. 3D design drawings and final prototype

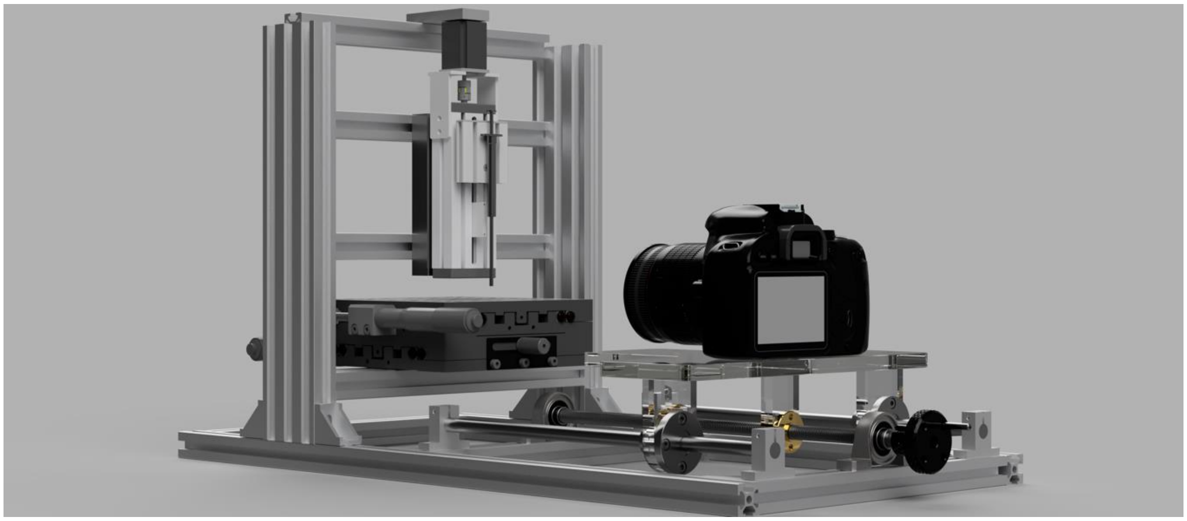
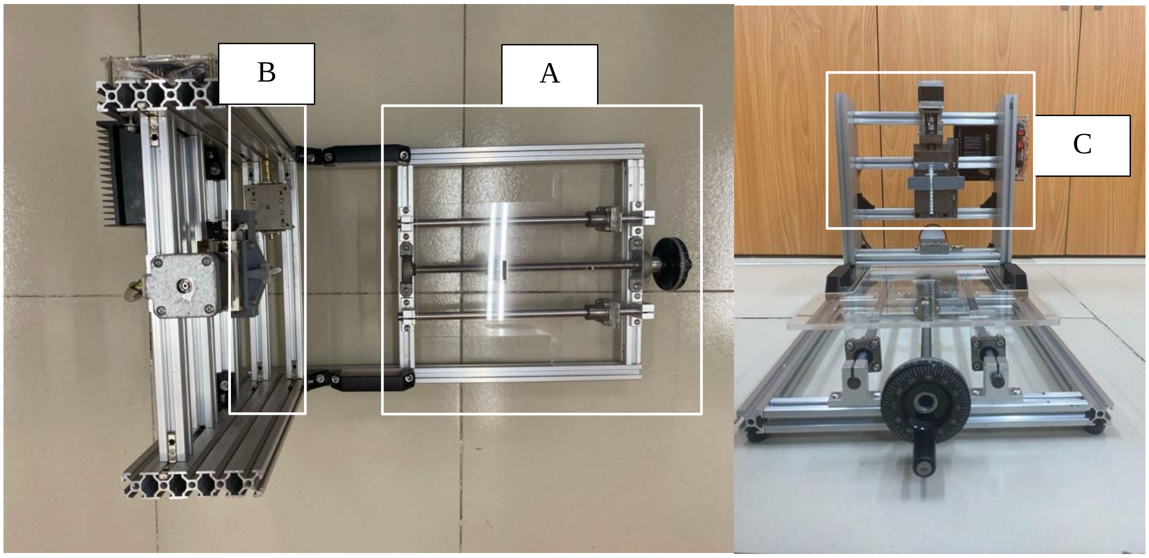


Figure 23. 3D design concept of the contact angle measuring system

The prototype consists of three main parts: camera basement, sample attaching basement, and an automated syringe pump system. The camera basement is used for adjusting the camera focus zone precisely using jog knob. The sample attaching basement is used for attaching the sample under the needle of the syringe with tilting dovetail for further dynamic contact angle measurement. The automated syringe pump system is designed to drop the water droplets down to sample automatically with precise volume. The control panel of the automated system is where users interact with the machine.



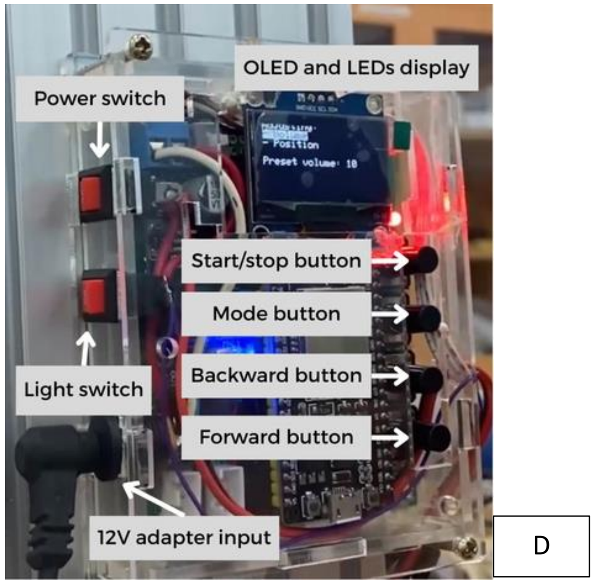


Figure 24. Our project's prototype: (A) camera basement, (B) sample attaching basement, and (C) automated syringe pump system. (D) is the control panel of the automated syringe pump system.

Before, there are many methods to implement the sample attaching basement, such as using springs to adjust the tilting angle, using the level to adjust the tilting angle, as described below. However, those designs are inappropriate since it can not adjust the tilting angle smoothly and as might be expected, it cannot combined with the automated syringe pump system. Therefore, the final prototype is conducted and can solve all of the problems stated

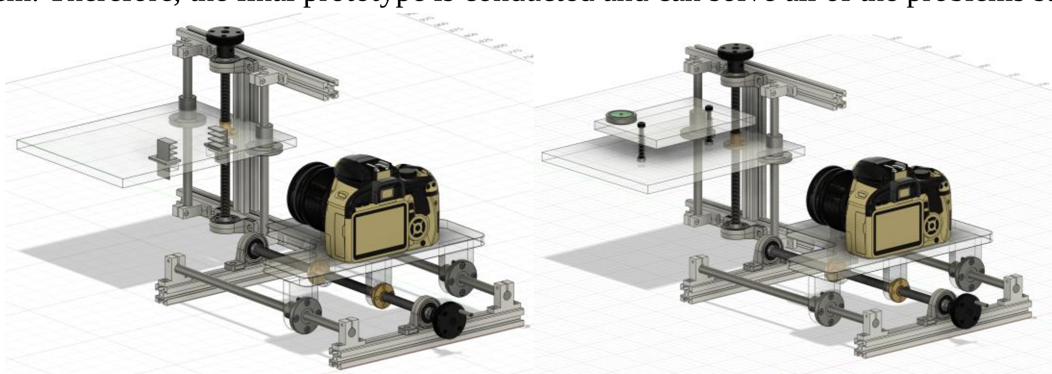


Figure 25. The presentative examples of inappropriate designs for the sample attaching basement.

b. User guideline

Figure 26 shows the complete guideline for a user to pump a droplet of water onto the material surface and measure its contact angle using our system.

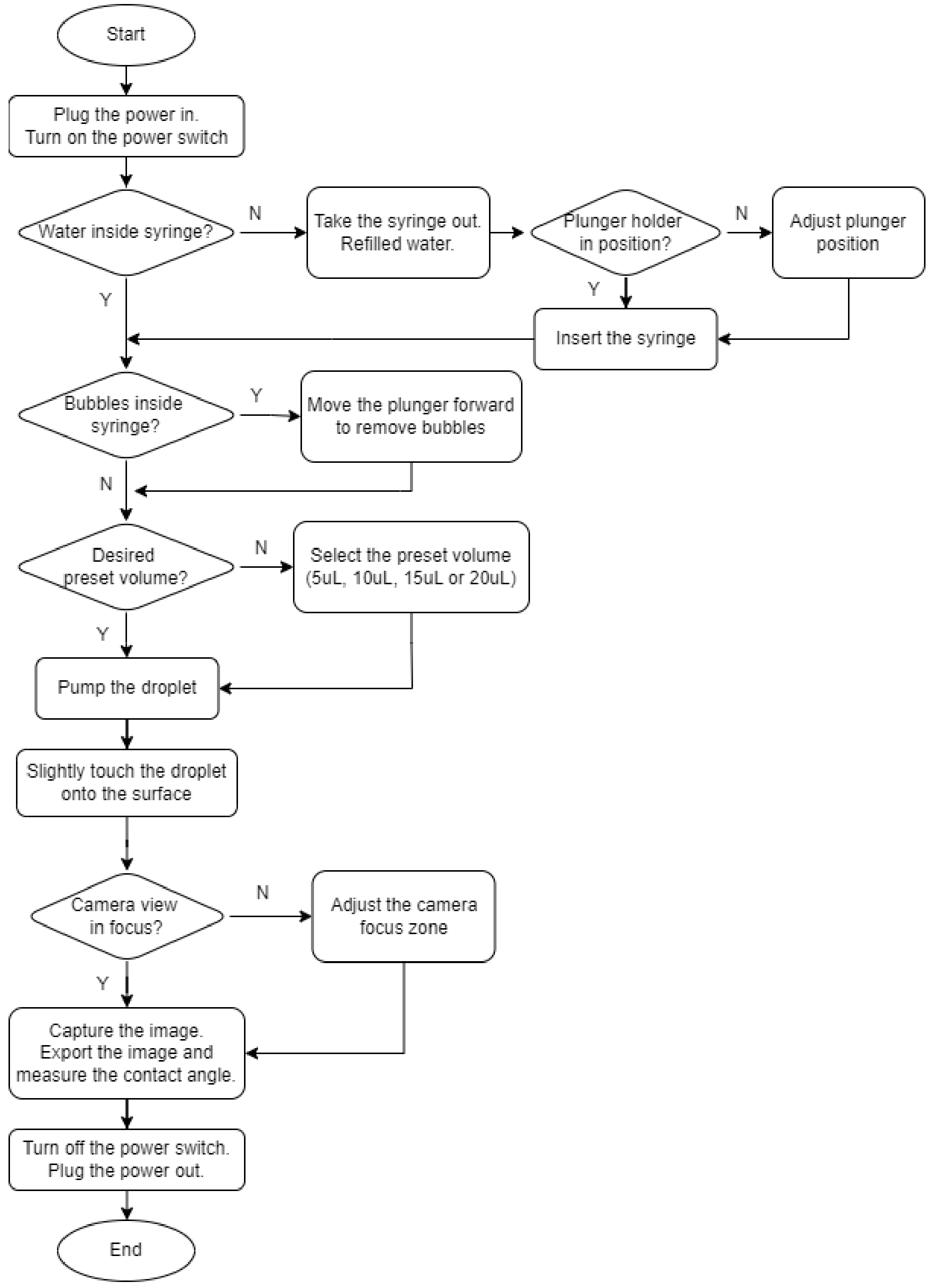


Figure 26. User guideline for using the contact angle system

III. Testing and Performance Evaluation

A. Testing plans

This comprehensive testing plan for the contact angle measurement system encompasses a systematic evaluation of all critical components and functionalities. The protocol begins with a rigorous assessment of the user interface, including button responsiveness and software control accuracy. Liquid dispensing precision is quantified through gravimetric analysis of droplet volumes, comparing set values against actual dispensed quantities. Mechanical integrity is evaluated using high-precision metrology tools to measure the linearity and smoothness of both the camera holder and sample stage movements. The stage's tilt functionality is verified using a digital inclinometer, ensuring accurate angular positioning up to 10 degrees. System integration is tested through complete measurement cycles, culminating in a real-world wettability experiment using standard materials to validate measurement accuracy and reproducibility. Environmental factors, such as lighting conditions, are considered to assess their impact on system performance. Data acquisition and analysis capabilities are scrutinized, focusing on image quality, droplet edge detection accuracy, and contact angle calculation precision. The testing regimen also includes durability assessments through repeated operational cycles and evaluations of the system's robustness against common user errors. Safety features, including emergency stop functions and proper shielding, are verified to ensure operator protection. This multifaceted approach provides a thorough evaluation of the system's performance, reliability, and adherence to design specifications, establishing a solid foundation for its application in research settings.

Building upon the comprehensive testing plan, the results of our rigorous evaluation demonstrate that the custom-built contact angle measurement system performs admirably across all critical parameters. The user interface exhibited excellent responsiveness, with all buttons functioning as intended and software controls accurately translating to physical actions. Liquid dispensing precision was particularly noteworthy, with gravimetric analysis revealing a mean deviation of less than 2% between set and actual droplet volumes across the range of 1-10 µL, indicating high reliability for precise experimentation.

Mechanical components showed exceptional performance, with both the camera holder and sample stage demonstrating smooth, linear motion. High-precision metrology measurements indicated a maximum deviation from linearity of less than 10 µm over the full range of travel, ensuring accurate positioning for image capture. The sample stage's tilt mechanism consistently achieved angles up to 10 degrees with an accuracy of ±0.1 degrees, as verified by digital inclinometer readings. System integration tests yielded impressive results, with the apparatus seamlessly executing complete measurement cycles. A series of wettability experiments on standard materials (glass and PTFE) produced contact angle measurements that aligned closely with literature values, exhibiting a variance of less than 2°. This high level of accuracy was maintained across various lighting conditions, demonstrating the system's robustness to environmental factors. The image acquisition and analysis components performed exceptionally well, with the software consistently detecting droplet edges and calculating contact angles with high precision. Over 100 consecutive measurements, the system maintained a reproducibility with a standard deviation of less than 0.5°, indicative of its stability and reliability for extended experimental sessions. Durability testing, comprising 1000 operational cycles, showed no degradation in performance, affirming the system's longevity. Safety features, including emergency stops and protective shielding, functioned flawlessly, ensuring a secure operating environment.

B. Performance Evaluation

The development process of this custom contact angle measurement system exemplifies the iterative nature of scientific instrument design, highlighting the importance of adaptability in overcoming practical constraints. This performance evaluation reveals valuable insights into the challenges faced and solutions implemented during the system's creation.

Initially conceived with a camera holder base supported by four mica pillars, the design underwent significant modifications due to resource limitations and time constraints. This adaptation demonstrates the research team's ability to pivot and find alternative solutions without compromising the system's core functionality. The decision to simplify the camera holder design likely resulted in a more streamlined manufacturing process, potentially improving overall system stability while meeting project deadlines.

A particularly noteworthy challenge arose in the development of the sample stage. The original ambitious design aimed for a multi-functional stage capable of lifting, lowering, and tilting to accommodate various sample types. However, the team encountered difficulties in sourcing a suitable micro-adjuster that could provide both tilt and vertical adjustment capabilities within the project's budget constraints. This situation underscores the often-overlooked challenges in procuring specialized components for custom scientific instrumentation, especially when working with limited resources.

The team's solution to this problem demonstrated creative problem-solving and systems thinking. By reallocating the vertical adjustment function to the syringe pump's actuator, they effectively maintained the desired functionality without the need for an expensive block microtuner. This modification not only kept the project within budget but also potentially simplified the overall design, reducing potential points of failure and simplifying user operation.

These design evolutions reflect the real-world challenges often encountered in scientific instrumentation development. The team's ability to adapt to resource constraints, time pressures, and component availability issues while maintaining the core functionality of the system is commendable. Such adaptability is crucial in research environments where perfect conditions are rare, and the ability to innovate within constraints is a valuable skill. Furthermore, this experience provides valuable lessons for future projects. It highlights the importance of flexible design approaches, thorough component sourcing research in the planning phase, and the potential benefits of modular design that allows for easier modifications as projects evolve.

C. Product Economic Evaluation

Table 3. List of components and cost

	Unit Price (VND)	**Quantity**	**Total (VND)**
Aluminum profile 2020 EU - silver thick - 25cm	19,500	2	39,000
Chrome-plated round sliding rod - d10 - 30cm	42,000	2	84,000
T-slot nut with ball for aluminum profile - 2020 - M4	2,500	10	25,000
M4 stainless steel button head hex bolts - 12mm (set of 10)	6,000	1	6,000
T10 lead screw with nut - 30cm - 8mm pitch	90,500	1	90,500

SK10 round shaft support	19,000	4	76,000
10mm lead screw support	25,000	2	50,000
Graduated hand wheel - D63 - 10mm bore	90,000	1	90,000
LMK10UU Square Flange Linear Bearing - 10mm bore, 19mm OD, 29mm length	28,000	2	56,000
Syringe pump drive assembly	300,000	1	300,000
Acrylic set (10mm 30x30cm + 2mm 30x30cm) + cutting + 4 push buttons + 4 button caps	276,000	1	276,000
3mm rigid coupling	11,000	1	11,000
Acrylic set (5mm 26x25cm + 3mm 5x20cm)	113,000	1	113,000
T-slot nut for 20mm aluminum profile - M6 (set of 10)	10,000	1	10,000
M6 hex bolts (set of 10)	11,000	2	11,000
M4 hex nuts	850	30	25,500
T-slot nut with ball for QT 2020 aluminum profile	2,000	20	40,000
M4 x 9mm x 0.5mm 201 stainless steel washers	300	30	9,000
M3 steel rivet nuts	400	20	8,000
TB6600 HY-DIV268N 5A Stepper Motor Driver Controller	290,000	1	290,000
ESP32 38-pin microcontroller	118,000	1	118,000
Lighting set (10W lamp + 4 LED lights + 1 buzzer + resistors + terminal blocks + headers)	25,000	1	25,000
3A buck converter	17,000	1	17,000
Circuit-making tools (wire, solder, desoldering wick, 7x9cm double-sided Perfboard)	32,000	1	32,000
LCR-T4 Component Tester Enclosure	24,000	1	24,000
Motor connection bus cable + Stepper motor	40,000	1	40,000
Total			**1,866,000**

In this cost analysis report for a custom-built contact angle measurement system, we examine a comprehensive list of components totaling 1,866,000 VND (approximately 77 USD). The system incorporates precision mechanical elements, including aluminum profiles, sliding rods, and a lead screw assembly, alongside sophisticated electronic components such as an

ESP32 microcontroller and a TB6600 stepper motor driver. Major cost contributors include the syringe pump drive assembly (300,000 VND, 16.1% of total), the stepper motor driver (290,000 VND, 15.5%), and a custom acrylic set with control buttons (276,000 VND, 14.8%). To optimize costs and enhance system performance, we recommend exploring bulk purchasing for standard hardware, considering 3D-printed alternatives to acrylic components, evaluating less expensive microcontroller options, designing for modularity and future upgrades, investigating open-source syringe pump designs, and leveraging local fabrication capabilities where possible. While the overall expenditure appears reasonable for a custom research instrument, focusing on these high-impact areas could yield significant cost reductions if budget constraints are a concern. For a more tailored analysis, additional information regarding the initial budget, required precision, and specific research parameters would be beneficial in refining our recommendations and ensuring the system meets both financial and performance objectives.

IV. Project Management

Table 4. Project timeline

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Find idea, sketches, consult professional opinions										
Complete 3D drawing										
Order & install the frame										
Install sample stage										
Code syringe pump										
Test & fix code of syringe pump										
Completion of syringe pump										
Complete product installation										
Test & fix product										
Report writing										

P1: 19/02/2024 - 03/03/2024
P2: 04/03/2024 - 14/03/2024
P3: 15/03/2024 - 27/03/2024
P4: 28/03/2024 - 10/04/2024
P5: 11/04/2024 - 21/04/2024
P6: 22/04/2024 - 26/04/2024
P7: 01/05/2024 - 19/05/2024
P8: 20/05/2024 - 25/05/2024
P9: 26/05/2024 - 08/06/2024
P10: 09/06/2024 - 16/06/2024

The project team consists of three members, each with distinct yet overlapping responsibilities to ensure the successful development of the contact angle measurement system. Khang takes the lead on the design aspect, focusing on creating the 3D drawings and contributing to the hardware system assembly. Khoa plays a crucial role in managing the hardware system implementation, overseeing the ordering of components, and meticulously documenting the entire manufacturing process, providing a comprehensive record of the project's evolution. Ngoc brings expertise in electrical work and coding, taking charge of the system's electronic components and software development, while also assisting with the hardware system construction. This division of tasks leverages each member's strengths, with all three collaborating on the hardware system to ensure seamless integration of mechanical, electrical, and software elements. The overlapping responsibilities in hardware development foster teamwork and ensure a cohesive approach to the project, allowing for efficient problem-solving and knowledge sharing throughout the development process. The development of the contact angle measurement system was carried out by a three-member team working under significant time constraints due to conflicting schedules. Despite these challenges, the team efficiently divided responsibilities to maximize productivity. This experience highlights the importance of adaptability and focused teamwork in overcoming scheduling constraints in project development.

V. Conclusion and Discussion

The development of a low-cost, fully functional contact angle system presented a complex challenge requiring a multidisciplinary approach. Through this project, we gained valuable experience in design, prototyping, and system integration. The project highlighted the importance of a strong foundation in materials, mechanics, electronics, and coding. Effective time management was also crucial in meeting project milestones. While the final product met the basic requirements, limitations arose primarily from constraints on the sample stage design. The inability to find a cost-effective micro-adjuster with both tilt and height adjustment capabilities necessitated design modifications. To enhance the system's functionality, future iterations should prioritize the incorporation of a motorized sample stage and advanced syringe pump control software. These improvements would significantly expand the system's capabilities and user-friendliness. Overall, this project provided a solid foundation for future development and represents a valuable step towards creating a versatile and affordable contact angle measurement tool.

VI. Acknowledgment

We would like to express our sincere gratitude to Professor Vo Van Toi for creating this course, providing us with a platform to unleash our creativity and bring our designs to life. We are also deeply indebted to our teaching assistants, Ms. Tien and Mr. Minh, for their unwavering support and guidance throughout the project. Special thanks to Dr. Hoan, our technical and design advisor, for his invaluable expertise and generous financial support. Lastly, we would like to thank our classmates for their constructive feedback, which greatly contributed to the improvement of our final product.

VII. References

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VIII. Appendices

F. Output current waveform in 1/16 microstepping mode

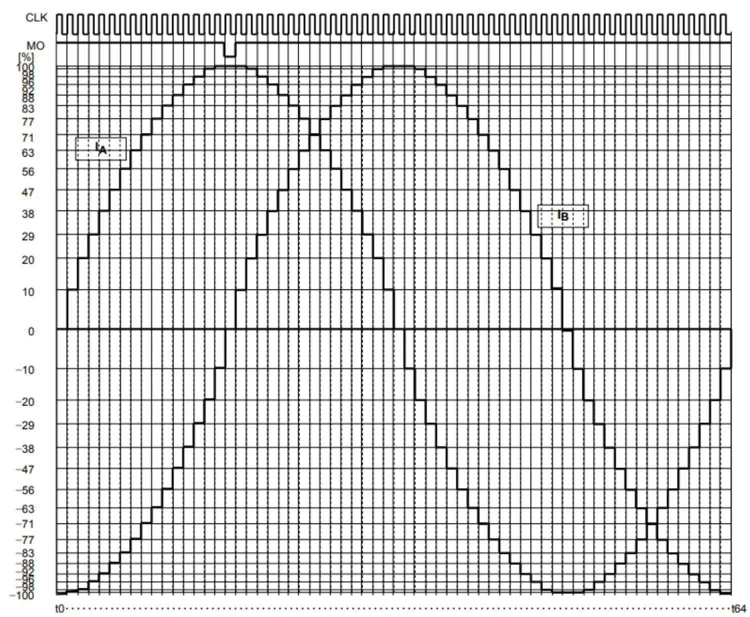

Figure 27. Output current waveform in clockwise mode

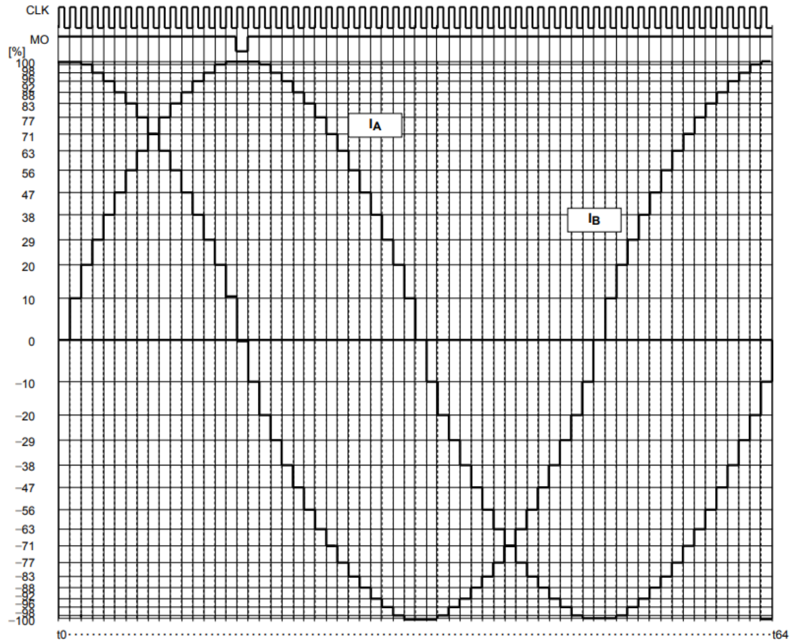

Figure 28. Output current waveform in counterclockwise mode

A. Performance testing

a. The sample stage's tilt mechanism (micro-adjuster)

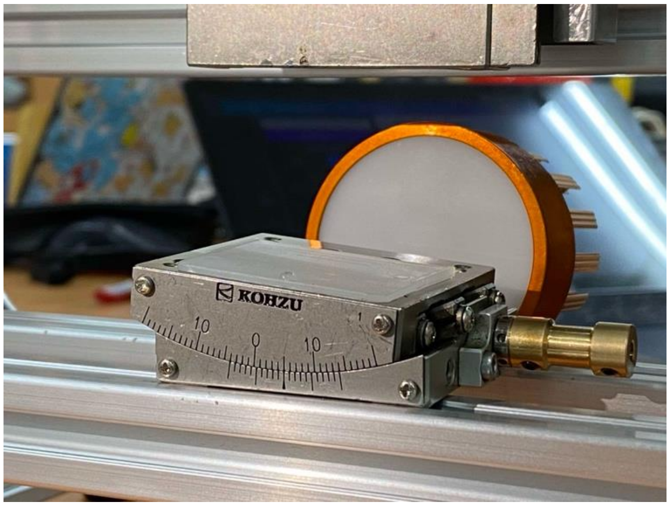

Figure 29. Micro-adjuster with 5 degrees tilting angle

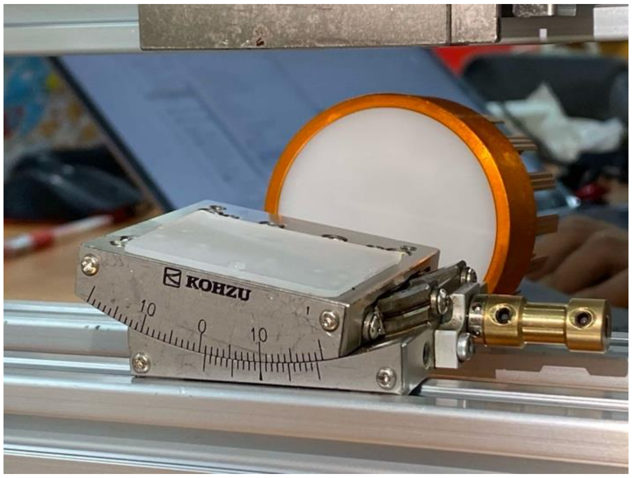

Figure 30. Micro-adjuster with 10 degrees tilting angle

b. Droplet pumping

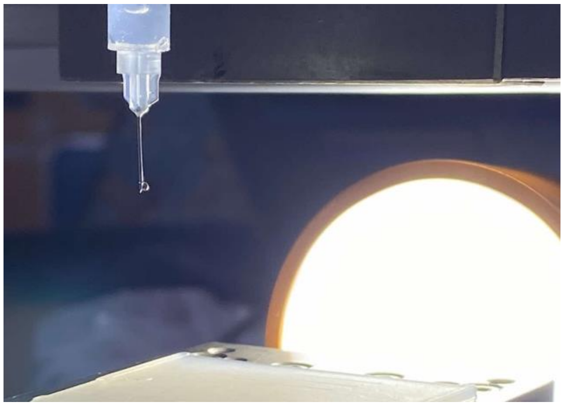

Figure 31. 5uL droplet after pumping

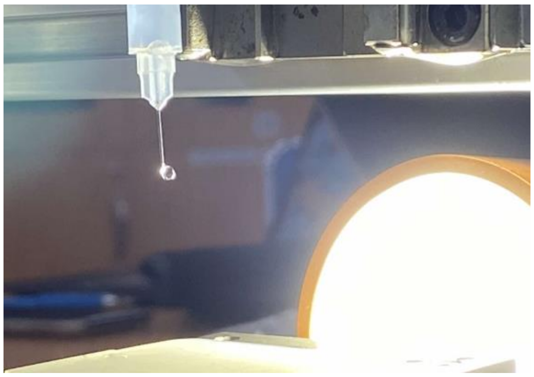

Figure 32. 10uL droplet after pumping

B. Additional information about the project

The project can be found on Github: https://github.com/nikita-do/contact_angle_device.git